

MEMORANDUM

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Project: Safer Streets Priority Finder – Phase 2

Re: Task 3: Segment Level Crash Risk Estimation Model Process

This memo documents the methodology for segment-level crash risk estimation, which translates block group-level fatal crash estimates into segment-level predictions of both fatal and non-fatal crashes. The framework then applies a Bayesian updating procedure to refine these predictions using recent, locally observed crash data. The workflow consists of three major steps: (1) Block Group Crash Disaggregation, (2) Non-Fatal Crash Estimation, and (3) Bayesian Updating.

Note: If the input study dataset includes only fatal crashes – such as when relying exclusively on FARS data - the model does not produce non-fatal crash estimates. In such cases, the Bayesian updating step would treat observed non-fatal crashes as zero, which biases the posterior estimates and renders non-fatal crash predictions unreliable.

Step 1: Block Group Crash Disaggregation

The objective of this step is to disaggregate the block group-level fatal crash estimates to individual roadway segments. This objective is achieved by the following 4 sub-steps:

Step 1.1 Assign Roadway to Block Group

Input Data and Parameters:

- Roadway Network: short segments prepared for sliding-window analysis. The default short segment length is 0.1 miles
- Block Group Estimates: block group level 5-year fatal crash estimation by modes, polygon layer.
- Buffer Width: buffer size of the influence radius of the roadway when assigning to block groups. The default value is 15 meters.

High Level Process:

The assignment of roadway segments to census block groups is driven by their spatial relationships. The method outlined in this memo also accounts for two key edge cases:

- Roadways that span several block groups or run very close to a neighboring block-group boundary.
- Roadways that are situated along shorelines or adjacent to areas that are not part of any block group.

When a roadway crosses multiple block groups, its mileage must be apportioned among them according to the length contained in each. Occasionally, however, a segment follows a boundary yet technically lies within only one block group (e.g., segment R1 in Figure 1). Because crashes on such a segment are influenced by conditions on both sides of the line, it is reasonable to attribute the segment to both block groups.

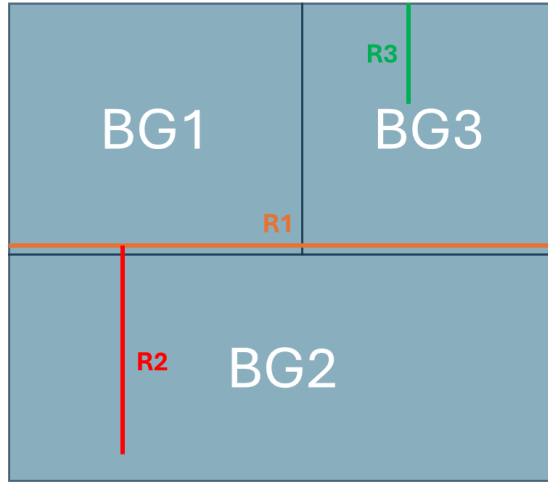


Figure 1. Example Roadway and Block Group Layout

To address these edge cases, we create a buffer around each roadway segment that serves as its “area of influence” (Figure 2). The overlap between this buffer and the surrounding block groups is then used to calculate the segment’s **effective mileage** within every intersecting block group. The effective mileage is computed as follows:

$$Effective\ Mileage_x = \frac{Overlap\ Area_x}{Total\ Overlap\ Area}$$

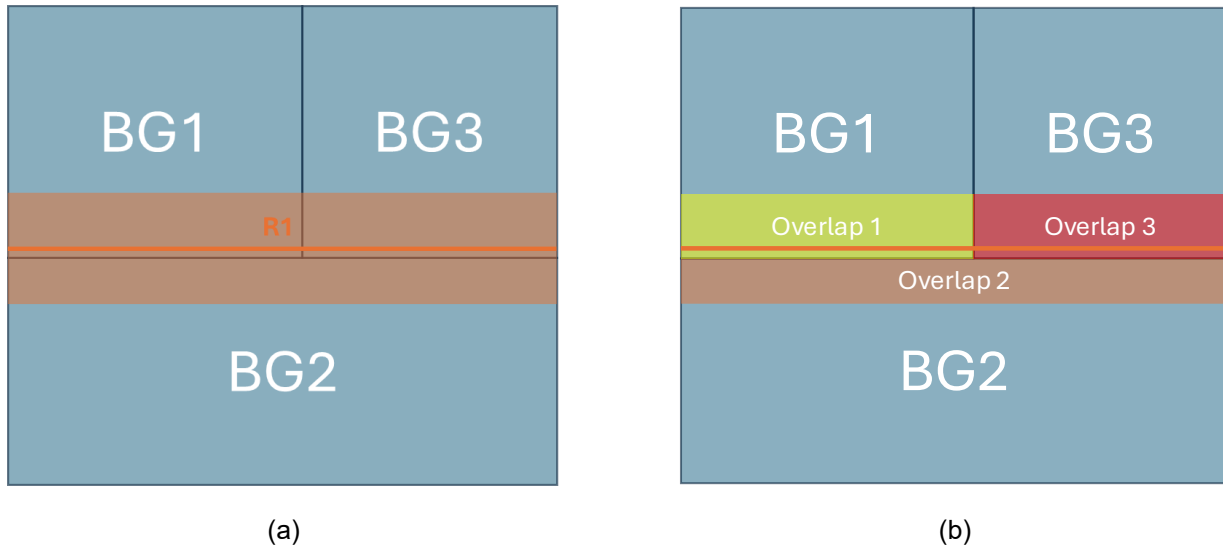


Figure 2. Roadway Assignment and "Area of Influence"

When a segment runs beside open water or land that is outside any block-group boundary, part of its buffer extends into “no-group” space. To ensure every foot of roadway is still assigned, we first clip away the portion of the buffer that lies outside all block groups (Figure 3). The trimmed buffer is then used to apportion the segment’s effective mileage, preventing any loss in the total mileage allocated to block groups.

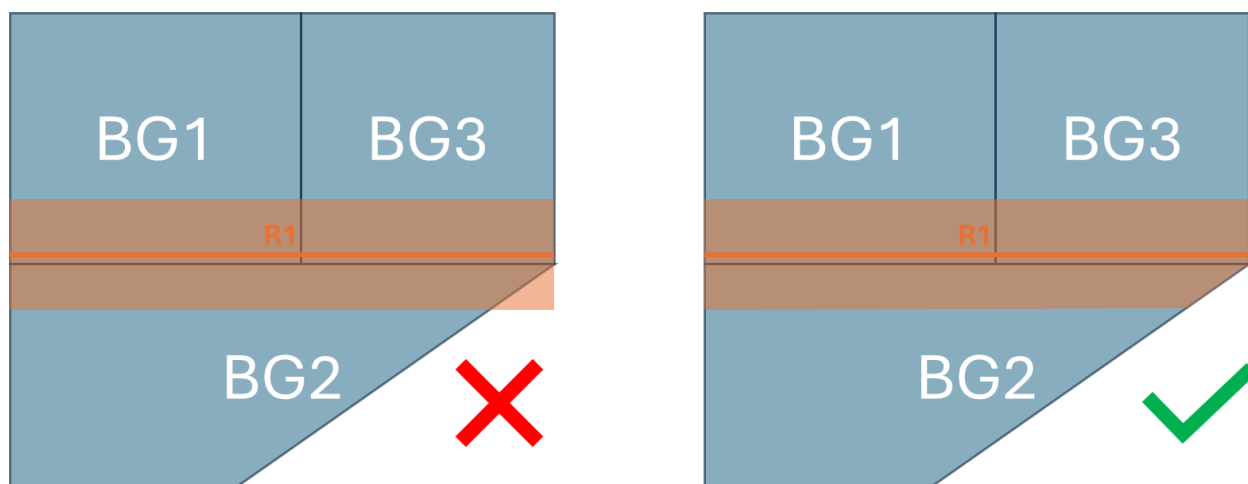


Figure 3. Example of Roadway Buffer Near Block Group Edge

Output:

- (Temp) Joined roadway and block group table: many-to-many records of the assignment of roadway and block group data with effective mileage.

Step 1.2 Compute Initial Fatal Crash Estimates

Input Data and Parameters:

- Joined Roadway and Block Group Table: output from [Step 1.1](#)
- Fatal Rate Lookup Table – Includes default annual fatal rates by mode and functional class for each state. The calculation of this default fatal rate lookup table is explained below.

Default Fatal Crash Rate Calculation:

For each state and year from 2018 through 2022, the average fatal crash rate per mile is calculated by dividing the number of fatal crashes in the state for each mode and functional class (from FARS) by the total mileage of roadway of that functional class for the state (from HPMS). The default fatal rate is then computed as the average of the annual rates across 2018-2022.

The functional class used by the model is slightly different from the “func_sys” codes used in HPMS. With “func_sys” codes 1 and 2 (“Interstate” and “Principal Arterial – Other Freeways and Expressway”) combined to represent the “Expressway” function class used in the model. Table 1 shows the mapping between the functional class used in the model and “func_sys” codes used in HPMS.

Table 1 Mapping between the functional class in SSPF and “func_sys” in HPMS

Functional Class In SSPF	“FUNC_SYS” Codes in HPMS
Expressway	1, 2
Major Arterial	3
Minor Arterial	4
Major Collector	5
Minor Collector	6

High Level Process:

This sub-step begins by attaching the default fatal-crash rate to the roadway–block-group table, matching on functional class. With the rate in place, it then calculates an initial fatal crash estimate for each record by multiplying the segment’s effective mileage by the corresponding fatal-crash rate.

Output:

- (Temp) Joined roadway and block group table with initial estimate: initial crash estimates are labeled as ``init_{mode}_k``

Step 1.3 Compute Scaled Fatal Crash Estimates*Input Data and Parameters:*

- Joined table with initial fatal crash estimates — output from [Step 1.2](#)

High Level Process:

This sub-step begins by summing the **initial** fatal crash estimates within each block group, separately for each travel mode (bicycle, pedestrian, and motor vehicle). For each block group it then derives a **scaling factor**, defined as the ratio of the block-group-level crash estimate to the summed initial estimates at the segment level for that block group. Since the block-group-level crash estimates were five-year estimates, and the initial estimates were annual-estimates, scaling factors were computed by first transforming block group-level five-year estimates to annual estimates, ($\text{scaling_factor} = \text{block_group_estimates_annual} / \sum \text{initial_estimates}$). Finally, every record’s initial estimate is multiplied by this factor, yielding a **scaled fatal-crash estimate** whose block-group segments sum exactly match the target total.

Because the joined table is at the roadway–block group pairing level, a single roadway segment may appear in multiple records (one per intersecting block group). In this step, ``est_{mode}_k`` represents the scaled estimate allocated to that specific segment–block group record, not yet the segment total.

Output:

- (Temp) joined roadway and block-group table with pair-level scaled estimates labeled ``est_{mode}_k``

Step 1.4 Aggregate Segment Level Fatal Estimates*Input Data and Parameters:*

- The joined roadway and block-group table with **scaled** fatal crash estimates (output from [Step 1.3](#)).
- The original roadway-network layer used in [Step 1.1](#).

High Level Process:

For every roadway segment in the source network, the procedure first aggregates the crash counts that were distributed across multiple block-group records and rolls the total back onto the segment itself. After the aggregation operation, all intermediate fields (initial estimate columns) were stripped away, retaining only the roadway-network primary key, the aggregated crash estimates ``est_{mode}_k``, one column for each mode.

Output:

- Fatal-estimates table at the roadway-segment level, containing the segment identifier plus fatal-crash estimates by mode.

Step 2: Non-Fatal Crash Estimation

The objective of this step is to translate the segment-level **fatal-only** predictions from [Step 1](#) into a full set of crash counts for all severities on the KABCO scale. This is achieved by the process described in sub-step 2.1.

Step 2.1 Compute Non-Fatal Crash Estimates

Input Data and Parameters:

- Segment level fatal-crash estimates `est_{mode}_k` produced in [Step 1.4](#).
- Severity-ratio lookup table that stores the typical ratio of A, B, C and O crashes to K crashes for each travel mode and state. Not all states' severity ratios were readily available, so states with missing ratios use the average known ratios from within the same Census Region¹

High Level Process:

For each mode, the procedure retrieves the appropriate severity ratios from the lookup table based on the state of the study area. The study area's state is determined by its centroid location. If the centroid falls outside of any state boundaries, it uses the state whose boundary intersects with that of the study area. If there are multiple intersecting states, the largest intersection area will be used as a tiebreaker. It then multiplies the segment's fatal-crash estimate by each ratio to derive the expected counts for A, B, C and O crashes:

$$\text{est}_{\text{mode}}_{\text{sev}} = \text{ratio}_{\text{mode}}_{\text{sev}} \times \text{est}_{\text{mode}}_k$$

Output:

- All-severity crash-estimate table at the segment level: New columns follow the pattern `est_{mode}_{sev}` for each severity, alongside the original roadway primary key.

Step 3: Bayesian Updating

The final step is to update the model-based crash predictions with evidence from recent, locally observed crashes. Using an Empirical Bayes (EB) framework, it balances the strength of the prior (Steps 1–2) against the observations, producing “adjusted” crash counts.

Step 3.1 Compute Crash Counts from Local Observations

Input Data and Parameters:

- Roadway Network: the roadway-network layer (with short window lengths) used in [Step 1.1](#).
- Local Crash Data: Crash data with mode and severity information.

¹ Census regions are one of the following – Northeast, South, Midwest, and West as shown in https://www2.census.gov/geo/pdfs/maps-data/maps/reference/us_regdiv.pdf

- Spatial Join Distance: max distance to join crashes to roadway segments. The default is 50 feet.

High Level Process:

Crash points are first spatially joined to any roadway segments within the max spatial join distance (default 15 meters). This assigns each crash to all segments within the search distance. Then, the number of crashes by mode and severity is tallied for each segment.

Output:

- *Local crash count*: a segment-level table of observed crashes by mode and severity.

Step 3.2 Compute Coefficient of Variance of Segment Level Crash Estimates

Input Data and Parameters:

- The original roadway-network layer used in [Step 1.1](#).
- Joined Roadway and Block Group Table: output from [Step 1.1](#)

High Level Approach:

The idea behind this Bayesian Updating approach is to use the coefficient of variance (CV) as a measure of confidence in the crash estimates. Since segment level crash estimates were influenced by nearby block groups, CV at block group level was used to estimate segment level CV. The segment level crash estimates CV were computed as the average of block group level CV weighted by the effective mileage of the segment within each block group.

Coefficient of Variance of Segment Level Crash Estimates:

The block group-level crash estimation model reports both block group estimates ($pred$), and the standard error of the estimates ($SE[pred]$), the CV of block group estimates can be computed using this expression:

$$CV[pred] = \frac{SE[pred]}{pred}$$

Since segment-level estimates may be influenced by multiple block groups, effective mileage of a segment within each block group (computed in Step 1.1) were used to compute the weighted segment-level CV from block group-level CV:

$$CV_s = \frac{\sum(Effective\ Mileage \cdot CV_g)}{\sum(Effective\ Mileage)}$$

Note that while the block group crash estimation model only estimates fatal crashes, since the lower severity estimates are scaled from fatal estimates, the coefficients of variance for fatal estimates also apply to other severities.

Step 3.3 Update Crash Estimates from Observations

Input Data and Parameters:

- All-severity estimates from [Step 2](#)
- Local crash count from [Step 3.1](#)

Bayesian Update Formulation

For each segment the adjusted crashes per year are expressed as:

$$N_{adjusted} = w \times N_{est} + (1 - w) \times N_{obs}$$

With

$$w = \frac{\alpha}{(1 + CV \cdot N_{est})^\beta}$$

Where,

N_{est}, N_{obs} is the estimated and observed crashes per year;

α is the maximum value of the weighting factor that will be assigned to N_{est} , (default = 0.8);

CV is the coefficient of variation of the N_{est} , indicating the confidence in the prior crash estimates;

β is the decay factor that controls how quickly the weight for N_{est} will reduce with the increase of CV and N_{est} , (default = 1).

The ideas behind this formulation are as follows:

- Using α to set a limit on how much observed crashes will influence the final “adjusted” estimate, regardless of the confidence in the prior crash estimates.
- Considering CV of crash estimates as confidence in the estimates.
- Using crash estimates as a measure of exposure to crash risks, assuming exposure increases as N_{est} increase. When exposure of the roadway is zero or low, this formulation reduces the weight of the observed crashes account for observations made under limited exposure.
- Using β as a decay factor to control how quickly the focus will be shifted towards observed crashes when significant exposure is present.

High-level process

1. Count observed average annual crashes by mode and severity for each segment.
2. Compute segment-specific weights w .
3. Blend predicted and observed counts to obtain posterior (“adjusted”) crashes.

Output

Updated Crash Estimates – posterior crash counts per segment, stored as $adj_mode_severity$.